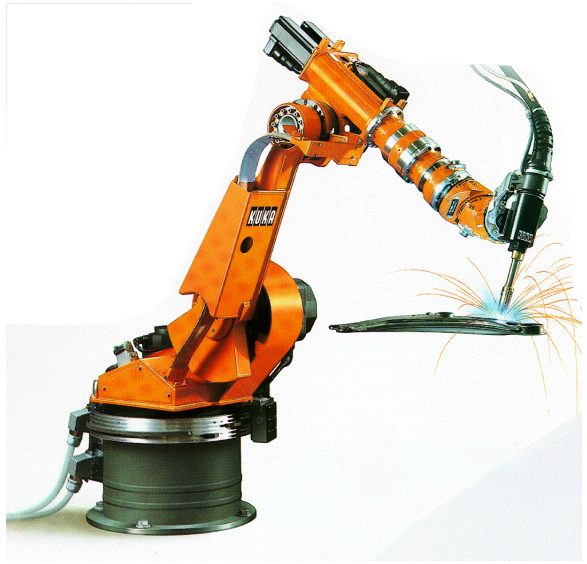
Introduction to Autonomous Robotics

Javier Gonzalez Jimenez

A definition of “Robot” [by Robot Institute of America]:

A robot is a reprogrammable, multifunctional **manipulator** designed to move material, parts, tools, or specialized devices through programmed motions for the performance of a variety of tasks.



This refers to **Industrial robots**, which are:

- fixed to the floor → can not change their locations (known)
- very precise and fast in their motion, but mostly repeat always the same thing
- working space is simple and perfectly known

We will not talk about this kind of robots!

We will cover here a more AI-related perspective of Robotics

Autonomous mobile robots

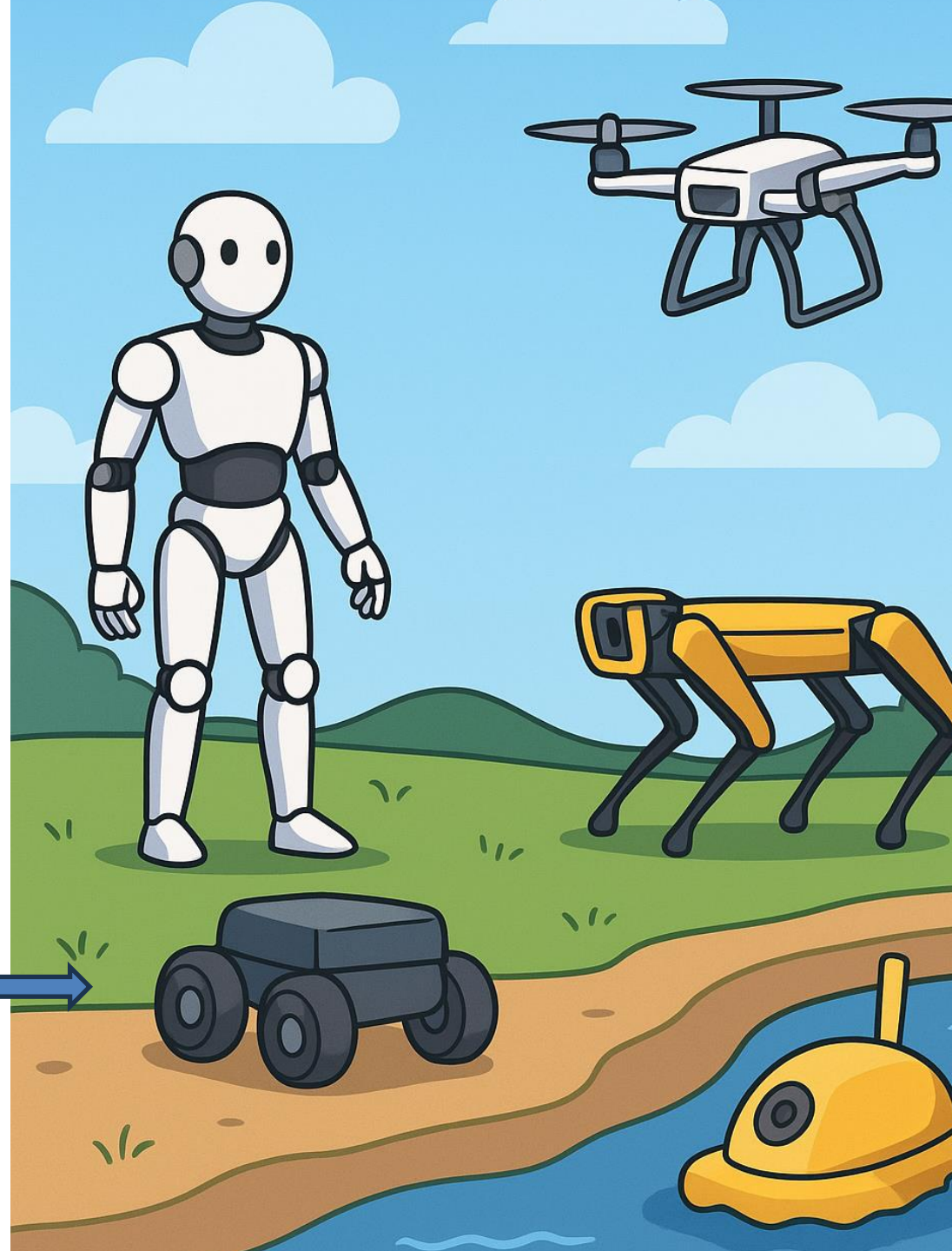


Should be able to …

- change their location in an environment where not everything is known a priori
- make intelligent decisions based on sensorial data

Types of Autonomous Robots
[according to type of locomotion]

Mostly, we will talk about
this type of locomotion



Paradigms in Robotics

Deliberative (Sense–Plan–Act)

- Robots build explicit **models of the world** using sensors.
- They rely on planning algorithms to compute optimal actions before execution.
- Limitations: computationally heavy, slow response in dynamic environment

Reactive

- Robots **directly map sensor inputs to actions** using rules or behaviors (biological inspiration)
- Advantages: fast, robust in dynamic and uncertain environments.
- Limitations: limited long-term planning, may struggle with complex tasks.

Hybrids

- Combines **high-level planning with low-level reactive control**.
- A **deliberative** layer handles goal setting and long-term decisions, while a **reactive** layer manages fast, real-time responses.
- Balances efficiency and adaptability, and is widely used in modern robotics.

This will be the focus of this course but incorporating uncertainty in the modeling, sensing and motion: Probabilistic Robotics

Probabilistic robotics

The key idea is **dealing with the uncertainty** inherent to sensing and motion, which propagates to

- the robot state (e.g. location),
- world representation (mapping) and
- planning and decision making.

Why

- Deal with inaccurate models of the world and the robot actions
- Deal with errors in sensors
- Robust to real-world changes (moving objects, unpredicted events, …)

Limitations

- Computationally demanding
- False assumptions about the probability distributions
- Approximate solutions

Key questions in robotics

Where am I?

- What's my position, within the places I already know? (Localization)
- What's in the workspace? Objects, landmarks, free-space (Mapping/SLAM)

How do I get there (safely)?

- How should I drive to avoid obstacles? (Obstacle avoidance)
- What is the best path to reach the goal? (Path planning)

How do I solve my mission?

(Task Planning)

AI and Robotics

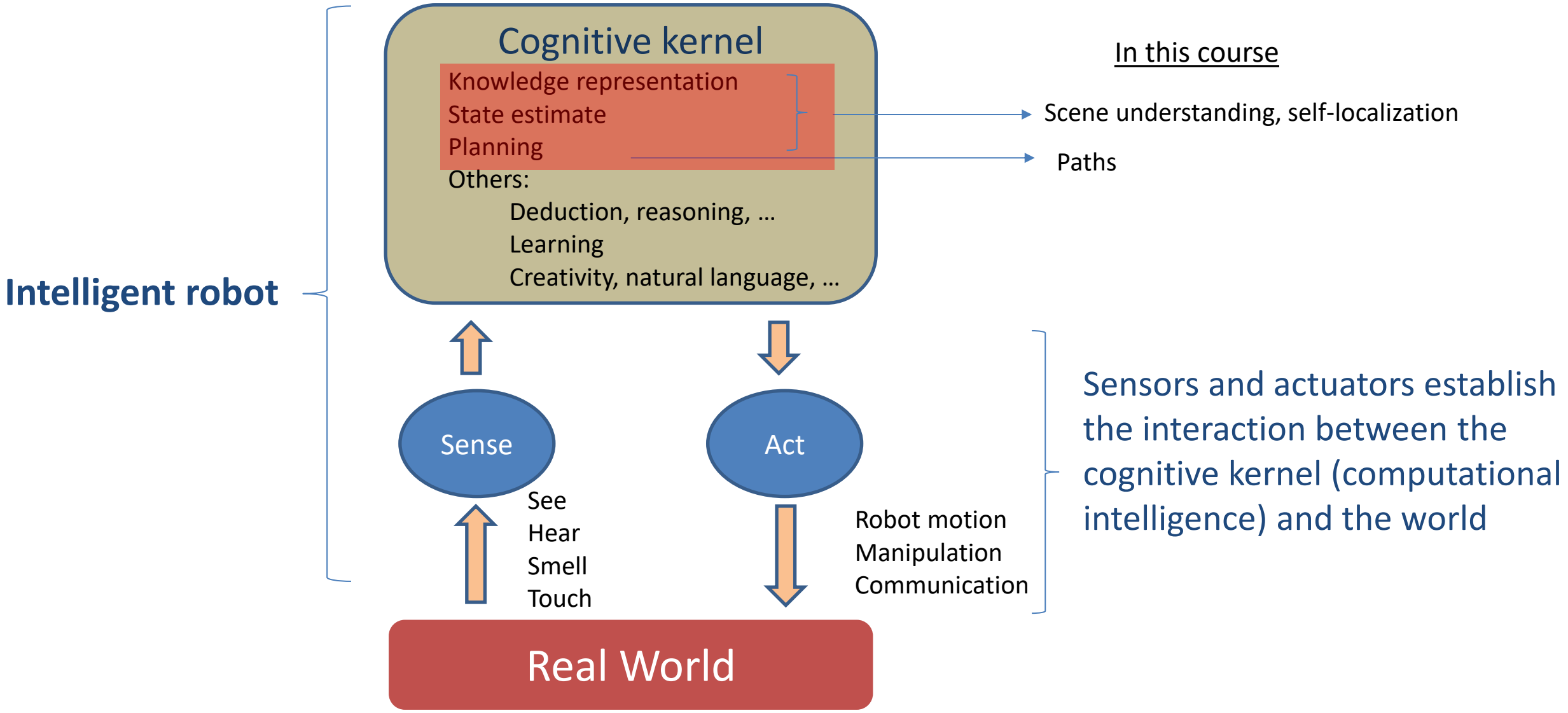
- An intelligent system needs "embodiment", just software is not enough for achieving truly intelligence
- **Embodiment hypothesis** (Linda Smith, 2005): "Intelligence emerges in the interaction of an agent with an environment and as a result of sensorimotor activity".

Embodiment entails both:

- Change the world (e.g. move, manipulate objects, talk, …) and
- Perceive the world

Robotics connects an AI system to the world using a **sensory-motor system**

AI and Robotics



Related disciplines

Robotics is an interdisciplinary field:

- Machine Learning
- Cognitive Science
- Computer Vision

} AI-related

- Software engineering
- Mechanical Engineering
- Control theory

++Mechanical engineering



++ Machine learning



++Control theory,



Applications of Mobile Robots














ColdFusion
https://www.youtube.com/watch?v=QwAYJPjW2gI

Syllabus

1. Introduction to Autonomous Robotics (this lecture)
2. Robot Control Architecture
3. Robot motion models
4. Robot sensors
5. Mobile Robot Localization
6. Mapping
7. SLAM
8. Motion planning



Evaluation

Two evaluation modalities:

- **Continuous:** for the grading in the 1st ordinary call
  - **Tests** (8 points)
    - Three partial exams (test questionnaires) covering both theory and practice
    - Grading of each of the partial exams. Required 3,5+/10
    - Final grade: Average of the 3 partial exams. Required 5+/10
  - **Practice** (2 points)
    - Attendance and timely submission of the practical sessions are mandatory
- **Non-Continuous:** for the grading in the 2nd ordinary call
  - Just one exam, covering theory and practical sessions (test questionnaires)
  - Submission of practical sessions (before the exam)